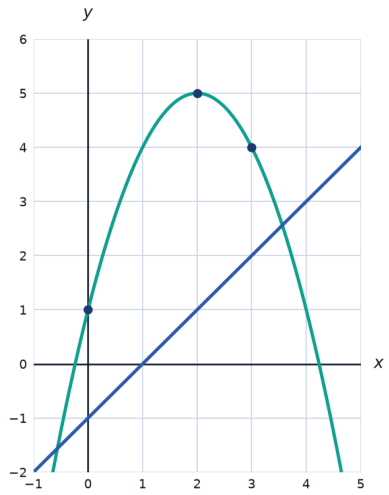
Exponential and logarithmic equations

Core Concepts: Exponential and logarithmic equations

1 For the model $f(x) = 3x + 7$ in exponential and logarithmic equations, evaluate $f(2)$.



a. 13

b. 16

c. 6

d. 17

2 Find the positive solutions to the quadratic equation $x^2 - 6x + 5 = 0$.

a. $x = 1$ and $x = 6$

b. $x = 1$ and $x = 5$

c. $x = -1$ and $x = -5$

d. $x = 6$ and $x = 5$

3 A data set recorded for exponential and logarithmic equations consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean?

a. 15.5

b. 16.5

c. 17.5

d. 19.5

Procedural Fluency & Algebraic Manipulation

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

4 For the model $f(x) = 3x + 7$ in exponential and logarithmic equations, evaluate $f(3)$.

a. 16

b. 19

c. 9

d. 20

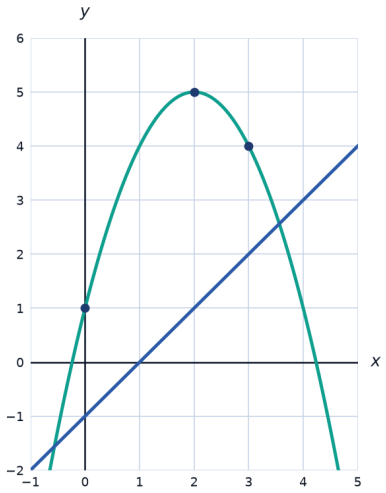
5 Find the positive solutions to the quadratic equation $x^2 - 8x + 12 = 0$.

a. $x = 2$ and $x = 7$

b. $x = 2$ and $x = 6$

c. $x = -2$ and $x = -6$

d. $x = 8$ and $x = 12$



6 A data set recorded for exponential and logarithmic equations consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean?

a. 15.5

b. 16.5

c. 17.5

d. 19.5

Graphical & Geometric Interpretation

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

7 For the model $f(x) = 3x + 7$ in exponential and logarithmic equations, evaluate $f(4)$.

a. 19

b. 22

c. 12

d. 23

8 Find the positive solutions to the quadratic equation $x^2 - 10x + 21 = 0$.

a. $x = 3$ and $x = 8$

b. $x = 3$ and $x = 7$

c. $x = -3$ and $x = -7$

d. $x = 10$ and $x = 21$

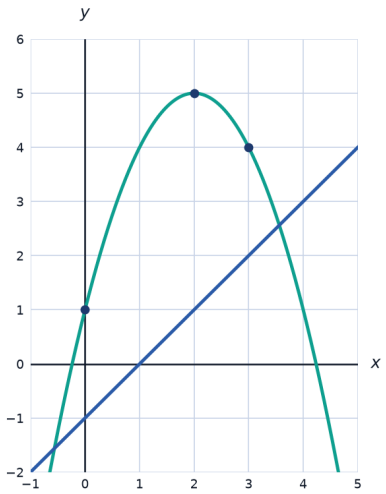
9 A data set recorded for exponential and logarithmic equations consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean?

a. 15.5

b. 16.5

c. 17.5

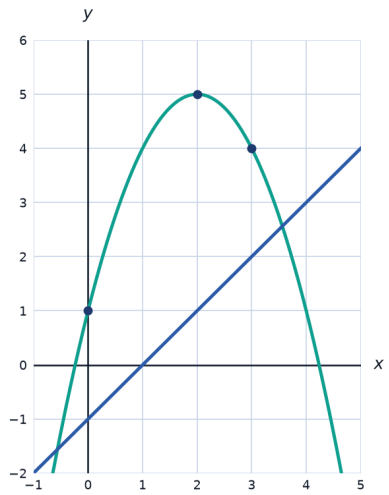
d. 19.5



Exam-Style Problem Solving & Modeling

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

10 For the model $f(x) = 3x + 7$ in exponential and logarithmic equations, evaluate $f(5)$.



a. 22

b. 25

c. 15

d. 26

11 Find the positive solutions to the quadratic equation $x^2 - 12x + 32 = 0$.

a. $x = 4$ and $x = 9$

b. $x = 4$ and $x = 8$

c. $x = -4$ and $x = -8$

d. $x = 12$ and $x = 32$

12 A data set recorded for exponential and logarithmic equations consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean?

a. 15.5

b. 16.5

c. 17.5

d. 19.5

© ClipSAT: By Mr. Mohamed Abdallah +966597688647